ABSTRACT

Access to written information remains a major challenge for visually impaired individuals, particularly when dealing with handwritten or non-digital content that requires manual transcription into Braille. The project presents BrailleGen, a low-cost Braille embosser designed to address this limitation by automatically converting both handwritten and digital inputs into tactile Braille output, with a primary focus on educational applications. The system supports multiple input formats such as images, PDFs, and PowerPoint files, and employs an AI-based text extraction approach using Google Gemini, with EasyOCR as a fallback to ensure reliability. The extracted content is translated into Grade 2 Unified English Braille (UEB) along with Nemeth code for representing mathematical and technical content, making it suitable for academic use. The processed Braille data is converted into binary form and transmitted to an Arduino-based hardware system that performs precise embossing using stepper and servo motors. The system achieves an overall accuracy of approximately 96%. By integrating intelligent software with cost-effective hardware, the system provides an efficient and affordable solution that enhances independent learning and accessibility for visually impaired users.

CONTENTS

LIST OF FIGURES

LIST OF TABLES

NOMENCLATURE

Abbreviation:

OCR – Optical Character Recognition

API - Application Programming Interface

UART - Universal Asynchronous Receiver-Transmitter

PDF- Portable Document Format

UEB - Unified English Braille

PPT - Powerpoint Presentation

PWM - Pulse Width Modulation

Electronic Components & Terms:

Arduino Mega: A microcontroller board based on the ATmega2560, used for controlling the embossers logic.

NEMA 17 Stepper Motor: High-torque motors used for precise positioning of the embossing head or paper feed.

DRV8825: A stepper motor driver carrier used to control the NEMA 17 motors.

SG90 Servo Motor: A lightweight, 180-degree rotation motor, likely used for the embossing pin mechanism.

CNC Shield: An expansion board (typically for Arduino) that simplifies the wiring for motor drivers.

IC 7805: A voltage regulator integrated circuit that provides a constant 5V output.

Chapter 1
INTRODUCTION

Access to written information is a fundamental component of education and communication. However, for people with visual impairments, accessing printed or written information is a challenging task. Most books, notes, and documents are prepared in a format that can be accessed visually. However, this becomes a challenge for people with visual impairments. Though technology has introduced many advanced tools to access printed information, it has not replaced the reading and writing process. Among these tools, Braille has been found to be an essential component in the development of literacy skills among visually impaired persons.

Braille is a tactile writing system consisting of raised dots in a matrix of six dots. Each pattern of raised dots corresponds to a character, number, or symbol. Braille can be read by touching the pattern with one's fingertips. Braille has become a widely accepted writing system in educational institutes, labeling schemes, and infrastructure. However, despite its significance, Braille materials are scarce. One of the significant factors that has contributed to the scarcity of Braille materials is the high cost and complexity of Braille embossers. Braille embossers are expensive equipment that are available only in specific institutes.

Another major problem is that most existing Braille embossers need digital texts as input. Therefore, when there is handwritten information, it needs to be manually typed before it can be converted to Braille. In real-life scenarios, handwritten information is quite prevalent, especially in classrooms. For instance, when there is a need to teach, the teacher writes on the board. Additionally, the students may have handwritten texts. Therefore, when there is a need to read such texts, it would be difficult for the visually impaired to read them.

It is worth noting that, due to recent advancements in artificial intelligence, it is now possible to develop systems that can read information from images. For instance, there is Optical Character Recognition (OCR) and other artificial intelligence techniques that can read both printed and handwritten texts. Therefore, there is an opportunity to develop systems that can convert both handwritten and digital texts to Braille.

Our project, “BrailleGen: A Low-Cost Braille Embosser,” seeks to address the challenges through the design and implementation of a low-cost and efficient solution. The system is designed to read various forms of input, including handwritten images, PDF files, and PowerPoint presentations. The system then utilizes AI tools to read the input, translate it to Braille, and finally print the output in Braille using a motor-controlled mechanism. The project is significant as it offers a solution to the challenges, providing accessibility to the visually impaired.

1.1 Overview

There are various challenges that visually impaired persons face in accessing written information. Even though digital technology has made various improvements in this field, there are still various limitations, especially when dealing with non-digital materials. Handwriting is a common practice in educational settings, and visually impaired persons usually rely on other people to read this material for them. This is not independent at all and may delay access to critical information.

Currently available braille embossing systems are mainly focused on digital input of texts. They need a person to type the text in a specific format before printing. This has made these systems inaccessible for converting other forms of materials into braille. These systems are also quite expensive and large in size, making them inaccessible to persons and organizations.

The other problem that has been identified in this field is the lack of integration of modern AI-based text recognition systems with braille printing systems. Even though there are various systems for reading image-based texts using OCR systems, these systems are not integrated with braille printing systems in a smooth way.

Therefore, there is a need for a system that can:

1. Accept multiple input formats, including handwritten contents
2. Automatically extract text using advanced AI techniques

3. Convert the extracted text into Braille format
4. Print the Braille output using a low-cost and efficient mechanism

1.2 OBJECTIVES

- Low-Cost Accessibility To design an affordable Braille embosser using off-the-shelf components, making tactile literacy accessible to those who cannot afford expensive commercial devices.

- Multi-Format Input Processing To create a versatile system capable of converting various sources including handwritten notes, digital images, PDFs, and PowerPoint presentations into Braille.

- High-Accuracy Content Extraction To ensure precise text retrieval by integrating Gemini AI and EasyOCR, allowing the system to accurately interpret both complex handwriting and standard digital text.

- Standardized Braille Translation To adhere to global literacy standards by implementing UEB for general text and Nemeth Code for technical or mathematical content.

- Efficient Hardware Execution To develop a reliable mechanical system using an Arduino controller, stepper motors for precise positioning, and servo motors for consistent tactile punching.

- Improved System Usability To provide a seamless, user-friendly workflow that allows visually impaired individuals to easily obtain Braille content from diverse real-world applications.

In general, the objectives of this project are related to improving accessibility, accuracy, affordability, and usability of this system.

1.3 SCOPE AND LIMITATIONS

The scope of the project includes creating a system that is complete and incorporates various processes such as input processing, text extraction, Braille conversion, and hardware-based embossing. The system is useful in various ways, especially in that it is able to handle both handwritten and digital data, thus ensuring its applicability in a variety of situations.

In relation to its scope, this system is useful in various places, such as in learning institutions where students and lecturers can use it to convert notes and other learning materials, respectively, to Braille, in libraries where people can use it to generate texts, and in personal use by visually impaired people to help them read various documents.

The system is also useful in that it incorporates various Braille formats, such as Grade 2 UEB and Nemeth code, thus ensuring that it is useful in various situations, such as reading and learning, respectively. It is also useful in that it incorporates constraints such as line length and page size, thus ensuring that it is readable by anyone who uses it.

However, there are a number of limitations that are associated with this system. The accuracy of the text extraction process relies on the quality of the image that is input into the system. The quality of lighting and the handwriting of the user may limit the functionality of this system. Furthermore, the use of the Gemini program may limit its functionality if there is no connection to the internet.

The other limitation that is associated with this system is the speed of the embossing process. The use of mechanical components in this system limits its speed as compared to other commercially available systems for the blind. There may be limitations in handling very large documents due to processing time.

Overall, this system has been able to provide a solution to a number of challenges that are associated with the blind population.

1.4 KEY INNOVATIONS

The system has a number of innovative features that set it apart from the traditional Braille embossers. One of the key innovative features of the proposed system is the inclusion of the AI-based text extraction process with the help of the Gemini tool. The Gemini tool is different from the traditional OCR tool in that it is able to understand complex handwriting and context.

Another innovative feature of the proposed system is the inclusion of the fallback approach with the help of the EasyOCR tool. This increases the reliability of the system by providing a backup approach for the extraction process in case the first approach does not work properly.

The system also provides a simple and effective method for handling different input formats by converting them to images and then processing them in a single flow. The use of Grade 2 UEB and Nemeth code is another important feature of the system. This feature makes it possible for the system to work with both normal and mathematical content. This is important in a school setting since both types of content are often needed.

In terms of hardware design, the use of the downward punching mechanism along with mirrored binary data is an important feature of the system. This makes it possible for the embossed output to be correctly aligned and easy to read.

The low-cost design of the system is another important feature of the design. This makes it possible for a larger number of people to use the system since it is easy to design and the components are readily available.

Chapter 2
LITERATURE REVIEW

Innovations in assistive technology are reshaping the lifestyle of people with visual impairment by simplifying the access and comprehension of information. Contemporary devices are capable of converting any printed or handwritten material into spoken language or tactile Braille thus users can read and learn on their own. By employing AI, smart software, and cheap hardware, the research community is enabling quicker and more precise text-to-Braille translation and thus the blind community is becoming equally involved in education, work, and daily life.

1. Abdullah Farhan Siddiqui, Md Misbahuddin, "A Tool for Converting Text to Braille Script to Empower Readability for Blind People" (2024)

One of the major driving factors for the development of Braille writing machines is the need to assist visually impaired individuals to live independently without depending on others. Recent developments, such as those proposed by Siddiqui and Misbahuddin, demonstrate the possibility of developing hybrid systems using software algorithms and mechanical actuators such as servo motors to convert text into Braille format. In addition, modern Braille writing machines make use of artificial intelligence and machine learning algorithms and optical character recognition to enhance the accuracy of Braille conversion. Advanced models such as deep learning networks are also being developed to assist in handling complex documents such as multi-lingual text and mathematical expressions. Furthermore, web platforms and speech recognition are being used to make the learning environment more interactive. However, factors such as maintaining accuracy while handling images of varying quality, handling multi-lingual text, and making the system cost-effective are still major limitations. For the future, improvements can be made in the areas of speed and the ability to handle more languages and complex content. These systems can become even more efficient and available for use. Nevertheless, continuous research and development are being carried out to make Braille writing machines more accessible to visually impaired individuals.

2. Andrean Ignasius, Joerio Christo Chandra, Roberto Oscadinata, Derwin Suhartono, "Image Pre-Processing Effect on OCR's Performance for Image Conversion to Braille Unicode" (2023)

The research by Andrean Ignasius, Joerio Christo Chandra, Roberto Oscadinata, and Derwin Suhartono focuses on the critical role of image pre-processing methods in enhancing the performance of Optical Character Recognition (OCR) systems for text recognition from images to Braille Unicode. The paper emphasizes the essential role of upgrading the quality of the image by means of such methods as binarization, noise removal, skew correction, and contrast adjustment before OCR extraction. All of these steps are dedicated to improving text readability and allowing the OCR engine to recognize characters more effectively, especially when confronted with different and deteriorated image inputs. Subsequently, the writers give more instances of how improved OCR can be utilized for the accurate transcription of text in Braille Unicode to produce reliable tactile outputs for visually impaired users. Their work complements present studies in the field of enhancing algorithms for conversion and hardware integrations that enable seamless interaction between printed or electronic text and Braille forms. Further, the paper reiterates that the trustworthiness of the text extraction depends heavily on the accuracy of the pre-processing stage, which has a direct impact on the success of the subsequent Braille Unicode translation. The experimental results shown in the paper reveal that there are considerable enhancements in OCR precision with pre-processing optimization, thus confirming the method's potential for real-life application. Consequently, the paper extends the research efforts in creating more dependable OCR-to-Braille systems, which in turn will be the source of reliable and scalable assistive technologies for the visually impaired and blind communities.

3. Falgoon Sen Apu, Fatema Islam Joyti, Md Ala uddin Anik, Md Wasi Uddin Zobayer, Atanu Kumar Dey, Sakib Sakhawat, "Text and Voice to Braille Translator for Blind People" (2021)

This study presents a different apparatus that is able to produce Braille from the input of text and voice for the blind people. In this way, the system merges text processing with speech recognition in such a way that the data can either be typed or spoken and the device will recognize Braille characters which it will also output to a single refreshable Braille cell. The device supports a maximum of 30 units to be connected simultaneously via Bluetooth or USB and thus, a teacher can efficiently guide several visually impaired students at the same time. Moreover, the system is characterized by a very low rate of errors in character recognition, the processing is done in real-time, it is cheap, and its handling is easy, thus, it is an optimal solution for the educational sector. Hence, it was designed with a focus on portability and user-friendliness, thus, the typical problems of standard Braille devices were overcome. What is more, the multi device connectivity feature of the program enhances collaboration as well as the communication among the blind community. Additionally, the feature of voice-to-text is another way of interaction that the users can avail of, particularly for those who have different abilities and preferences. The research highlights the potential of such merged solutions to raise literacy levels and educational attainments among the blind by, at the same time, providing them with more communication tools, which are also more versatile and accessible. Essentially, this paper helps to extend the range of the assistive technologies and therefore, it is a step forward towards inclusive education for the visually impaired community.

4. Sangeeta Kumari, Akshay Akole, Pallavi Angnani, Yash Bhamare, Zaid Naikwadi, "Enhanced braille display use of OCR and solenoid to improve text to braille conversion"(2020)

An innovative system has been put together that essentially improves the performance of a Braille display by merging Optical Character Recognition (OCR) technology with solenoid actuators to elevate text-to-Braille efficiency and accuracy. The system fetches digital text, makes use of OCR to get the characters, and in a physical way, takes solenoid-controlled motion to Braille the characters. The employment of solenoids allows for an output that is dynamic and refreshable Braille in which the issues of portability and real-time text conversion are solved. The invention has several benefits in that it is smaller in size, consumes less power, and has quicker response times than a typical Braille display. Besides, the system's modularity opens up possibilities for the next generation of the device such as the use of more miniaturized solenoids and advanced controllers in order to further improve the performance. Also, this technology goes a long way in solving the problem of high prices that are usually associated with refreshable Braille displays, thus, making such an assistive technology accessible to a larger number of people. This approach is a giant leap in the field of assistive technology that gives the visually impaired user community convenient, efficient, and easy ways to read digital text.

Table 1 – Text-to-Braille vs OCR: A Comparative Analysis

Paper & Year	Key Features	Limitations	Methods we adopt
Arroyo et al, 2020	Low-cost Braille embosser using printer head and solenoids	High cost, slow printing, limited to Grade 1 Braille	Adopted printer head mechanism
Siddiqui et al, 2024	Affordable, text-to-Braille, compact and user-friendly	Cost and input limitations	Adopted hardware & control techniques
Ignasius et al, 2023	OCR-based Braille embossing, image processing	Accuracy and mechanical complexity	Adopted OCR methods
Apu et al, 2021	Text/voice-to-Braille, low-cost single Braille cell	High cost, complex design	Adopted software techniques
Kumari et al, 2020	OCR with solenoid tech, portable Braille system	Cost and digital input dependency	Adopted image preprocessing methods

Chapter 3

TECHNOLOGY IN GENERAL

The proposed Braille embosser system has been developed through the integration of various technologies, both in the software and hardware domains. These technologies have been chosen in such a manner that the process of the Braille embosser system is done in the best way, starting from the input process to the final output. The Braille embosser system can process different types of inputs and then convert them into the required output in the form of Braille, with the help of the integration of artificial intelligence, programming tools, and motor control mechanisms.

3.1 SOFTWARE TECHNOLOGIES

The software part of this system plays an important part in making all this run smoothly. It deals with different types of input, such as images, PDFs, or presentations, and converts them into the required form for Braille printing. First, it detects the input type, followed by the extraction of the input content using the appropriate method. Then, the content is cleaned and processed to remove any unnecessary characters before it is converted into Braille form. In addition to this, the software also takes care of the overall flow of the process, ensuring that each step takes place in the correct order without any problems arising. It also takes care of the communication between the software and hardware, so the data can be transferred appropriately. In this entire process, care is taken to ensure the accuracy of the output, as even the slightest mistake can impact the overall output. Also, some error handling is incorporated into the system, such as handling low-quality input or the failure of the content extraction part. The software part of this system needs to be simple, accurate, and efficient in dealing with different real-world situations.

3.1.1 Google Gemini

Google Gemini is used as the main technology for extracting the text in the system. It is a multimodal model of artificial intelligence that can work with both images and text. In this system, it is used to extract the text from handwritten images, scanned documents, PDF files, and presentation slides. Unlike other optical character recognition techniques that make use of pattern recognition techniques to recognize characters in images, Google Gemini uses advanced contextual understanding to recognize characters more effectively.

The use of Google Gemini in the system helps to avoid errors in the system while extracting the text. This is because the system can work effectively even if the images are not clear. It can also work effectively in cases where there are different fonts and styles. This improves the reliability of the system as a whole because the quality of the Braille output depends greatly on the accuracy of the extracted text.

3.1.2 Programming Language (Python)

The programming language used for the software part of the system is Python. It is the programming environment in which all the processing activities are done. The file inputs are processed, and the Gemini API is accessed to perform the required operations on the extracted text, which is then prepared for transmission to the hardware part of the system.

One of the major reasons for choosing Python as the programming language for the project is its simplicity, which makes the programming task easier and more efficient. It also supports the library functions of the API, which can be helpful for implementing the project using artificial intelligence for the text extraction process. It works as a bridge between the data inputs and the hardware system by controlling the entire data flow.

3.1.3 Serial Communication

Serial communication is used for the transfer of data from the software system to the hardware system. In the project, UART communication is used for the transfer of the processed Braille data to the Arduino hardware.

The transfer of data takes the form of characters or strings over a serial port at a specific baud rate, usually 9600 bps. This ensures the transfer of data in a synchronized manner, i.e., without any errors. Serial communication is a critical part of the project as it acts as a bridge between the software processing unit and the hardware system for embossing the Braille cells. Without proper communication, the hardware system will not be able to obtain the proper instructions for embossing the Braille cells.

3.2 HARDWARE TECHNOLOGIES

The hardware part of this system is responsible for executing the physical actions that need to take place in order for the printing of the Braille text to happen. The data that is processed by the software is taken and converted into action. Different parts of the system work together in order to ensure that the dots that form the Braille text are clear and well-positioned. The microcontroller is one of the most important parts of the system, as it receives the data and sends the signals. The stepper motors are the ones that move the print head and the paper, while the servo motors move the dots. The coordination of these parts is important in order to ensure that the output is accurate and consistent.

The performance of the system is important, as it will not only influence the speed of the system, but the quality of the embossing will also depend on the selection and overall composition of the system. The overall aim of the system is to ensure that the final output is clear and easy to read.

3.2.1 Arduino Mega 2560

The Arduino Mega 2560 acts as the main control unit in the system. It receives the processed text data from the software and converts it into control signals that are necessary for operating the motors. It coordinates and controls all the hardware units in the system, including stepper motors and servo motors.

The Arduino Mega processes the data and decides the sequence of actions that need to be performed in order to print a Braille character. It also generates step signals for stepper motors and PWM signals for servo motors. The Arduino Mega has a number of input/output pins and memory capacity, making it suitable for controlling more than one component in the system.

3.2.2 Stepper Motor

Stepper motors are used for the purpose of achieving precision in the movement of the system. In the project, NEMA 17 Stepper Motor motors are used for the purpose of moving the printing head in the horizontal direction and controlling the movement of the paper.
Stepper motors are capable of moving in steps, i.e., the motor can rotate in steps, and each step can rotate the motor by a specific angle, which enables the motor to move with precision.
The precision of the stepper motor is very useful in maintaining the gap between the Braille characters and the lines correctly, and the movement of the embossing head to the exact location for the purpose of printing the dots accurately is ensured by the stepper motor.

3.2.3 Servo Motor

The system uses a servo motor for its embossing mechanism. The type of servo motor used for this system is the SG90 Micro Servo Motor. This type of servo motor controls the punching action that makes the Braille dots on the paper.

The servo motor works based on Pulse Width Modulation (PWM) signals. This signal allows the servo motor to rotate to a specific angle. In this system, the servo rotates to a specific angle to press the paper and then returns to its original position. This type of mechanism helps to create Braille dots on the paper. The use of a servo motor helps to achieve high precision.

3.2.4 Motor Driver Module (DRV8825)

The motor driver modules are used for the control of stepper motors. The Arduino board on its own is not able to provide the necessary current to drive the stepper motors. The motor driver modules, therefore, are used as an interface between the Arduino board and the stepper motors.

The motor driver receives the step pulses and direction signals from the Arduino board. The signals are then converted into the necessary current and voltage for the operation of the stepper motors. The motor driver also supports microstepping, which allows the stepper motors to rotate smoothly. The driver also has the necessary protection circuits, which protect the stepper motors from damage.

3.2.5 CNC Shield

The CNC Shield is used as an interface between the Arduino Mega 2560 and multiple stepper motor drivers such as DRV8825, and is mounted directly on top of the Arduino to create a compact and organized setup. It provides dedicated slots and clearly labeled connections for step, direction, and enable signals, which simplifies the overall wiring and reduces the chances of connection errors. By eliminating complex manual wiring, the CNC Shield improves system reliability and makes assembly and maintenance easier. It also enables the control of multiple stepper motors simultaneously, allowing coordinated movement of the carriage and paper feed mechanisms. Additionally, it supports microstepping configuration for smoother and more precise motor operation. Overall, the CNC Shield helps in organizing the hardware, improving performance, and making the system more scalable and efficient.

3.2.6 Integration of Technologies

The system works by integrating all the above mentioned technologies. The software will be responsible for the processing of the input and the extraction of the text. The hardware will be responsible for the actual printing. The extracted text will be translated into Braille and sent to the Arduino using serial communication.The Arduino will be responsible for interpreting the data and controlling the motors. The stepper motor will be used for movement, and the servo motor will be used for embossing. This will ensure that the Braille characters are printed correctly. The integration of the software and hardware is essential for the proper functioning of the system.

Chapter 4
METHODOLOGY

The methodology of the proposed system describes the overall workflow that has been followed while designing the system for the development of the Braille embosser. In this system, software and hardware components are integrated in a structured and systematic way. The process starts with the acquisition of the input data, which may be images, PDFs, or presentation files. After the acquisition of the input data, the text is extracted from the data using a specific technique. After the text has been extracted from the input data, it is processed and converted into Braille format. After the text has been converted into Braille format, it is represented in binary form so that it can be understood by the hardware components of the system. Finally, the data is sent to the hardware components of the system for the process of embossing. In this system, all the components are designed in a way that they can work independently of each other while also being integrated with other components of the system. This helps in identifying the errors of the system and also improves the performance of the system.

4.1 SOFTWARE DEVELOPMENT

The software of the proposed system for the Braille embosser plays a vital role in the handling of different types of inputs and converting them into Braille format for execution by the hardware. The system can handle different types of inputs, such as handwritten images, PDF files, and PowerPoint presentations. After the file is uploaded, the software recognizes the type of input and processes it. For images, the AI model is used for text recognition. For PDFs, the images are created first, and then the text is recognized. For PowerPoint, the text is directly extracted.However, if the main method of extracting text fails to do so successfully, an OCR method is used as an alternative to ensure successful text recognition. Once the text is extracted successfully, it is converted to Braille form using standardized methods such as Grade 2 Unified English Braille for normal texts and Nemeth Code for mathematical expressions. This has helped the system process the input efficiently and generate commands for the hardware for Braille embossing.

4.1.1 Implementation Strategy for OCR

The first step in the process of developing the software is text extraction. For this purpose, the system mainly uses the Gemini tool, as it is highly efficient in text extraction using its advanced AI features. With the help of Gemini, the system is able to efficiently handle both printed and handwritten text. If the user is asked to insert an image or a document consisting of images, the system uses the Gemini tool to process the image and perform text extraction on it.

But sometimes, the quality of the image or the handwriting may not allow the system to perform text extraction using the Gemini tool. Therefore, the system also uses an OCR-based approach for text extraction. If the text extraction using the Gemini tool is not efficient or is incomplete, the system uses an OCR-based approach to perform text extraction. This ensures that text extraction is not completely impaired.

4.1.2 Braille Translation Development

The text is then processed and converted into Braille format. The system uses standard Braille encoding to ensure accuracy and readability. For regular text, Grade 2 Unified English Braille (UEB) is employed. The Grade 2 Braille contains contractions, which minimize space. The contractions enable frequently used words or a combination of letters to be represented with fewer Braille cells.

For mathematical text, the system employs the Nemeth Code, which is developed to represent mathematical symbols and equations in Braille. The translation process involves the conversion of all characters, numbers, and symbols into their Braille equivalent. Proper spacing is also maintained to ensure that the text is readable by visually impaired users. The translation process is important because, without accuracy, the content can be misinterpreted.

4.1.3 Serial Communication

After converting the text into Braille, the system creates a binary form of the Braille cell. In the 6-dot Braille cell, each of the dots is represented in binary form. If the dot is there, it is represented as 1; if the dot is absent, it is represented as 0. Since the embossing mechanism works on the principle of downward punching, the binary form of the data is inverted before transmission. The processed binary form of the data is then sent to the Arduino using the serial communication method. In the serial communication method, the data is sent in a structured manner. In the proposed system, the binary form of the data represents a particular Braille character. The Arduino understands this data and processes it accordingly. Proper synchronization is maintained so that there is no delay in the process.

4.2 HARDWARE PROTOTYPING

In the hardware design, emphasis is given to the transformation of digital Braille information into physical embossed dots on the paper. The hardware design follows the concept of hardware prototyping, where the design is enhanced through continuous improvements. The system is implemented using mechanical components such as guide rods, lead screws, and motor mounts to ensure accurate and precise movements.

4.2.1 Mechanical Design Iteration

In the mechanical design, emphasis is given to the structure to ensure accurate movements along the horizontal and vertical axes. Guide rods are used to ensure stability, while lead screws ensure accurate linear movements. Several design iterations are performed to ensure accurate alignment, minimize vibrations, and ensure smooth movements. Particular emphasis is given to the alignment of the punching system to ensure proper Braille readability since any misalignment would compromise the readability of the Braille output. During each iteration, adjustments are made to ensure appropriate spacing between the Braille dots.

4.2.2 Motor Control Calibration

In the system, stepper motors and servo motors are utilized. These types of motors should be calibrated to ensure the accuracy of the system. Stepper motors are utilized to control the movement of the print head (x-axis) and the paper (y-axis). These stepper motors should be calibrated to ensure the accuracy of the movements.

For the Braille system, the servo motors should be calibrated to ensure the accuracy of the positioning of the punching pin to the top, middle, or bottom of the Braille cell. Additionally, the servo motors should be calibrated to ensure the accuracy of the punching action to be performed on the Braille cell.

4.2.3 Assembly and Integration Testing

Once the individual software components are designed, they are integrated to form a complete system. In addition, the software is integrated with the hardware components such as the motors and controllers. During this process, integration testing is done to ensure that the commands from the software are being properly executed by the hardware.

The system is also tested to ensure that all movements and actions are taking place simultaneously. For instance, the X-axis motor is moved to the correct position, the servo is positioned correctly to punch the paper, and the punching action is also taking place at the correct time. This step ensures that everything is running smoothly and producing accurate results.

.

4.3 TESTING AND VALIDATION

Testing and validation are critical components of the development of the Braille embosser system, as they ensure that both the software and hardware are functioning correctly under different conditions. The Braille embosser system has different stages of processing, such as input processing, text extraction, Braille translation, binary conversion, and mechanical embossing. Testing the system also helps in identifying any possible errors or inefficiencies in these stages of processing. The Braille embosser system is tested using different types of input, such as handwritten images, PDF files, and PowerPoint files. These inputs are processed using the software, where the text is extracted and then converted into Braille using standard rules. The Braille information is then converted into binary and then passed to the microcontroller for embossing. While testing the software, aspects such as the accuracy of text extraction, accuracy of Braille translation, and accurate communication between various parts of the program are taken into consideration. On the other hand, while testing the hardware of the Braille printer, aspects such as the accuracy of the stepper motor, servo motor, and punching mechanism are taken into consideration. Moreover, the accuracy of Braille dot formation and their alignment is also monitored while testing the Braille printer. This is done through step-by-step testing of the Braille printer, and the accuracy of the Braille translation is ensured for the end users of the Braille printer.

4.3.1 Unit Testing

In the process of unit testing, each component of the system is individually tested to ensure that it is working properly. This involves checking the performance of the OCR in extracting the text, the performance of the text conversion to Braille, and the performance of the motor control. Each module of the system is individually tested to ensure proper performance before it is connected with other components of the system. For example, the performance of the OCR is compared with the original text to ensure proper accuracy. In the same way, the performance of the Braille translation is also compared with the standard Braille translation rules. The motor performance is also tested to ensure proper movement. This process helps in identifying the errors at an early stage. This process also ensures proper performance of each component of the system before moving to the next step.

4.3.2 Integration Testing Framework

After the unit testing of each module is done, the integration of all the components is performed and integration testing is done. This ensures whether all the parts of the system are working properly when connected together to perform the required tasks such as input processing, text extraction, Braille translation, binary creation, and hardware control. The entire process is checked step by step from giving input to the program in the form of a file to finally embossing the Braille output using the Arduino board. During this phase of testing, special emphasis is given to the coordination between the software and the hardware to ensure proper synchronization between the two. It also ensures that the data is properly transferred between the different stages without any loss or errors. Any discrepancies or delays in the process can be identified and corrected at this time.

4.3.3 Performance Benchmarking

Once all the modules have been individually tested, they are integrated into the system, which is then tested. This helps to ensure that all the software and hardware components integrated into the system are functioning properly. The entire system's functioning, starting from the input, text recognition, Braille conversion, and finally embossing, is verified step by step. The communication between the software and Arduino is also verified, ensuring that the data is sent appropriately without any delay or loss. This step also helps in identifying the problems related to the delays or the transmission of the data from one part of the system to another. It ensures the smooth functioning of the system, providing accurate Braille output. It also helps in identifying the small mistakes that might not be identified during the testing of the individual modules, such as small delays or movements. These issues can be addressed appropriately before the final implementation. The continuous testing also ensures the smooth functioning of the system during long-term usage. It also provides the confidence to ensure the system's ability to handle different types of input without failing, making the output accurate and reliable.

Chapter 5
SOFTWARE IMPLEMENTATION

The software of this Braille embosser project is one of the most important aspects of the entire system. This software helps in the proper working of the system. The software works by taking different types of inputs and converting them into a proper format that can be used to carry out Braille printing. Unlike other basic systems that only allow one type of input, this project allows the user to insert different types of inputs like handwritten images, PDF files, and PowerPoint presentations. Therefore, the software of this project has been designed in such a manner that it can handle each of these inputs in a proper manner without any confusion. The software of this project works by carrying out several operations like identifying the type of input, converting it if needed, extracting the text, formatting it, and translating it into Braille. All these operations have to be carried out in a proper sequence to ensure that the output of the system is clear. The system has also been designed in such a manner that it can handle small errors. Therefore, even if one of the operations of the system does not function properly, the entire process will still continue.

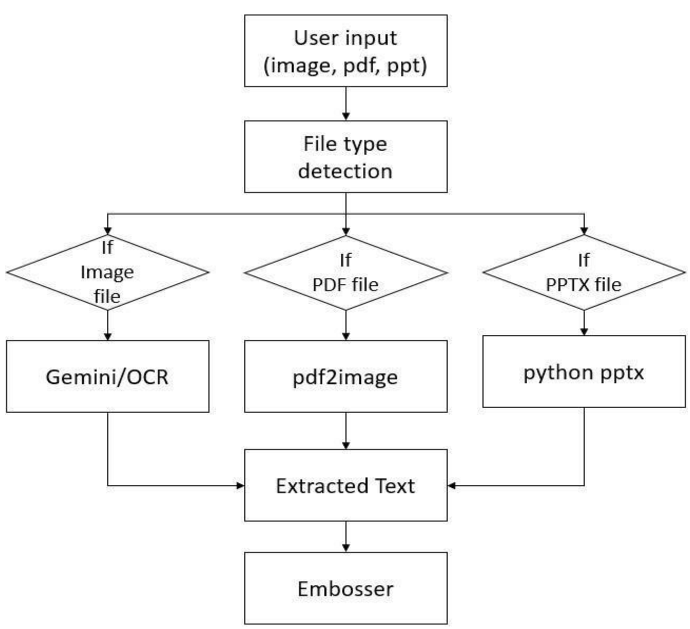

Fig 5.1: Flowchart of Input Processing and Text Extraction

The flowchart illustrates the overall working of the software component of the proposed Braille embosser system. The process starts with user input, which can be in the form of an image, a PDF file, or a PowerPoint file. The system starts with file type detection, which determines the nature of the input file.

Depending on the file type, the process takes different paths in the flowchart. When the file type is an image, the text from the image can be extracted using either the Gemini or OCR method. When the file type is a PDF file, the file is first converted into images using the pdf2image library, and then the text can be extracted from the images. When the file type is a PowerPoint file, the text can be directly extracted using the python-pptx library.

After extracting the text from the input file, the text from different file types is unified, and then the process goes to the embosser module for further processing for Braille conversion and printing.

5.1 TEXT EXTRACTION USING GOOGLE GEMINI

The process of extracting the text is a critical part of the proposed Braille embosser system, as it essentially bridges the gap between raw inputs and useful information, which can then be processed further. The proposed system can accept a wide range of inputs, including handwritten images, scanned documents, PDF, and presentation slides, among others. However, the inputs received by the system are not directly usable for the Braille conversion process unless they are processed to obtain the required information.

In the proposed project, Google Gemini is used to perform the task of extracting the required information from the inputs received by the system. The Google Gemini is a highly advanced multimodal artificial intelligence model that can understand visual as well as textual information. It can understand the context of the information, which is presented to it, as opposed to the traditional approaches that can only recognize the information but cannot understand the context of the information. Thus, the proposed system becomes more efficient and capable of handling real-world inputs with high accuracy.

5.1.1 Need for Gemini

The conventional approach of using OCR may face difficulties when dealing with real-life data, such as blurred images, varying light conditions, and handwritten data. This may cause incorrect character recognition, which can have a direct effect on the Braille conversion. In applications like this, where the end user may be visually impaired, this can have a substantial impact on the usability of the application.

In order to overcome these difficulties faced by the conventional approach, the application of Google Gemini has been incorporated. This can interpret the data more efficiently by using its advanced models of learning. This can help to improve the overall quality of the text obtained by processing the input data. This can ensure that even complex data can be handled more reliably by using the Gemini approach, resulting in a more accurate Braille conversion.

5.1.2 System Integration

In the overall system architecture, Gemini has been integrated as a core module in the system's software pipeline. Once the user has given the system the input file, the system will undergo the necessary preprocessing steps to improve the quality of the data.

After the images have been processed and improved in quality, the system will send them to the Gemini model using the API interface. The system will then process the images and send the processed output in a structured form. This output will be sent to the next module in the system's pipeline for further processing. In this module, the output will be converted into Braille patterns. The processed output will be sent to the Arduino Mega 2560 for further processing.

5.1.3 Working Methodology

The process of Gemini in the system involves a step-by-step procedure to correctly extract the text from the given input. The process begins by the user feeding the input to the system in the form of images, PDF, or a presentation file. However, the raw form of the input might be noisy or distorted, hence the need for preprocessing to make the data clear and readable.

The preprocessed data is then sent to the Gemini system via an API call. The Gemini system then processes the input data by employing its neural network to correctly identify the characters, words, and relationships within the given data. The Gemini system, therefore, does not only rely on the patterns of the characters but also the meaning of the characters.

The Gemini system then produces the extracted text, which is received by the system, and then the extracted text is passed to the Braille conversion module for the conversion of the text to a dot pattern. Therefore, the above procedure ensures that the extracted text is correct for the purpose of the system.

5.1.4 Features of Gemini

Google Gemini has a number of features that make it appropriate for the intended use. The first feature is its multimodal capacity, which enables the system to process images as well as text at the same time. This enables the system to process a wide range of inputs without the need for different processing approaches. The other feature of the system is its high accuracy in recognizing printed as well as handwritten texts. This is a critical feature, especially for the intended application, as the inputs might not always be well formatted. The system can also recognize complex inputs, including those with a mix of texts and graphical information, which can be difficult for the system to process. The system is also scalable, which enables it to be used for a wide range of inputs as well as for improvements in the future. These are some of the critical features of the Google Gemini system, which enhance its flexibility and functionality as a whole.

5.1.5 Advantages and Limitations

One of the advantages is that it improves the accuracy of text extraction. When compared to other OCR techniques, Gemini can comprehend the context of the text. This helps in identifying the text better. For instance, in handwritten notes and scanned documents, Gemini can identify the text better compared to other techniques. This is because Gemini can comprehend the context of the text. This helps in improving the accuracy of Braille conversion.

Another advantage is that Gemini can process different types of input data such as images, PDF files, and presentation slides. This is because Gemini can process different types of input data without different techniques for different types of input data. For instance, other techniques use different techniques for processing images and PDF files. However, Gemini can process all types of input data. Moreover, Gemini can process input data without heavy preprocessing. This is because Gemini can process input data even if it is not clean and properly structured.

However, there are certain limitations associated with the use of this model. One of the major limitations is that the model requires internet connectivity in order to function properly. This is because the model is accessed through the internet using the API. Therefore, the model will not function properly in offline conditions.

The performance of the model also depends on the quality of the input provided to the model. If the image is too blurry, then there might be certain errors in the text extracted.

Another limitation of the model is that it uses the API in order to function. This means that the performance of the model is dependent on the API. If there is any delay in the API, then the performance of the model will also be affected. This is the reason why the model might not function properly. It is evident that the use of the Gemini model has improved the text extraction process and has improved the performance of the system. However, there are certain limitations associated with the use of the model, and these are the areas where improvements can be made in the future.

5.2 INPUT PROCESSING

Input processing is the first step in the entire system. It is the step in which the system receives the input from the user and processes it for further steps. Since the system is designed to work with various types of inputs, this step needs to be able to recognize and process them accordingly.

When the user uploads the file, the system recognizes the file extension and checks what kind of file it is: an image, PDF, or PowerPoint. Recognizing the file type automatically makes it easier for the user, as there is no need to specify the type of processing to be carried out. After identifying the file, the system processes it according to the type.

The major aim of this step is to standardize all the inputs to one particular form that can be easily handled and processed by the text extraction module. As the system is designed to work mainly with images, all the inputs are converted to the image format before moving to the next step. Another important factor that needs to be considered during the input processing stage is validation. The system checks the validity of the file that was uploaded by the user. If the file is not valid or is corrupted in some manner, the system may send a notification to the user.

5.2.1 Image Input Handling

When the input is an image, it is processed directly by the system. There is no need to convert the image in any major way. However, when considering an image, the quality of the image plays a major role in ensuring the accuracy of the text. In such cases, some preprocessing steps may be required to improve the quality of the image. Some of the preprocessing steps may include adjusting brightness and contrast to improve image quality, removing noise to remove unwanted disturbances, sharpening the image to improve the quality of text, etc. This may be particularly helpful when extracting text from handwritten notes, as the style of writing may differ from person to person.

Once the image has been preprocessed, it can be fed into the text extraction module. Gemini is used as a primary tool to extract text from images. This tool can easily comprehend printed as well as handwritten text. One of the advantages of using Gemini is its ability to comprehend context. This helps in recognizing words even when the handwriting is not clear.

In some cases, the image may also have irregularities such as overlapping text, inconsistent spacing, and background images. These may affect the accuracy of the extracted information. However, the system is designed to accommodate these irregularities as much as possible. In case the main method for extraction does not give the desired results, the system also provides a fallback option.

5.2.2 PDF Processing

However, PDF files are handled in a slightly different manner. PDF files may be composed of multiple pages and may also have different kinds of content. PDF files may be composed entirely of digital text or may be scanned documents. In such a case, the system converts the PDF file to an image.

This process is carried out by making use of the pdf2image library. This library converts the PDF file to an image. The PDF file may be composed of multiple pages. The system treats each page as an individual input. This makes it easy to process the PDF file.

After the PDF file is converted to an image, it goes through the same process as the regular image input. The text from the PDF file or scanned document is extracted by making use of Gemini or EasyOCR. The extracted text from all the pages is combined to form a complete document.
While making use of multiple-page documents, the data needs to be handled in an efficient manner. The system maintains the sequence of the pages. This is done to make sure that the final output is in the correct sequence. This is especially important in the case of educational documents.

5.2.3 PPT Conversion

PowerPoint files are used in academic and professional settings, hence the importance of the system being able to handle them. Unlike PDF files, PPT files contain structured data, i.e., slides, text boxes, and graphical content.

The system uses the python-pptx library to process PPT files, which enables it to access the content of each slide in the file. The library extracts text from different elements, including titles, bullet points, and descriptions. This is a good feature, especially when dealing with structured data, as it maintains the logical structure of the content.

Once the text is extracted, it can be used as is or converted into an image format to conform to the rest of the system. In most cases, the text is cleaned and formatted before being passed to the next stage.

The major challenge when dealing with PPT files is the inclusion of non-text content, i.e., images and diagrams, which the system currently ignores as it is based on text content only. However, the inclusion of PPT files is a major advantage of the system.

5.3 SERIAL COMMUNICATION

After the text is converted into Braille and binary format, it needs to be sent to the hardware system for embossing. This communication is done with the help of a serial communication protocol between the processing unit and the Arduino microcontroller.

Serial communication is selected for the process due to its simplicity and reliability. The communication process involves the transmission of data in a serial manner, with a group of binary values representing a Braille cell. The communication process provides the Arduino with the instructions in the right order. For the process to be properly synchronized, it may involve the use of markers that show the start and end of the data sequence. This provides the Arduino with the ability to identify the start of a new page or line of the document. The process of synchronization is important for the proper working of the motors. The communication speed, also known as the baud rate, is selected according to the needs of the system. The baud rate provides a balance between the speed and reliability of the system for proper working without the loss of data. Once this data is received by the Arduino, it interprets the binary code and operates the motor accordingly. The interaction of software and hardware is very important for the proper functioning of this system.

5.4 ERROR HANDLING AND FALLBACK

In real-world applications, it may not always be possible to have the best possible conditions. The image may not be clear, the file may not be properly formatted, or the network may not be working properly for the AI tool to function properly. So, error handling is an important part of the software.

One of the important features of the software is the use of a fallback approach for extracting the text from the image. Although the primary tool for this is the Gemini tool, it may not always function properly. So, the EasyOCR tool is used as a fallback approach for the extraction of the text from the image.

Another important feature of the software is the handling of errors that may come while working with the file. Although the software can properly convert a PDF file into the required format, it may not always be the case. So, the file may be reprocessed by the software. Similarly, the software may skip the elements in a PPT file that it does not support while processing the file.

The other significant factor is the validation of the extracted text. The system checks if the extracted text is valid and not empty. If the results are not satisfactory, the user may be asked to input a more understandable string. By incorporating these error handling mechanisms in the system, it becomes more robust and reliable. The system ensures that the user gets the expected output even in adverse situations.

Chapter 6

HARDWARE IMPLEMENTATION

The hardware of the Braille embosser is the part of the device that actually does the work of producing the output. This is the part of the device that takes the information provided by the software and converts it into Braille dots on a piece of paper so that a visually impaired person can read it. The software does the work of extracting the information and converting it into Braille dots; however, the hardware makes sure that these dots are printed correctly and are well spaced apart.

The design of the hardware is a bit challenging since it has to move accurately and exert the correct amount of pressure while embossing the dots. Even a small amount of inaccuracy in the movement of the hardware or the amount of pressure it exerts while embossing the dots can affect the readability of the Braille dots produced by the device. Therefore, the hardware is set up correctly so that it does not produce any errors while working.

The hardware part of the system is composed of various mechanical parts, motors, and control circuits. It includes a frame to hold all the parts of the system, a mechanism to move the paper, and a punching mechanism to punch the Braille dots. Motors are used to control the movement of the print head and the paper. Each component of the system is designed for a specific function, and all the components must function properly to obtain the required output.

The cost is also an important factor of the system in this project. Most Braille embossers used commercially are expensive as they use special parts for their components. In this project, readily available parts such as stepper motors, servo motors, and an Arduino are used to build the system. This helps to keep the cost of the system low while ensuring good performance of the system.

The hardware is also designed with modularity in mind. This means that individual components can be replaced or upgraded without affecting the entire system. For example, motors can be replaced with higher-performance versions, or the punching mechanism can be improved for better durability. The hardware design has a focus on maintaining a good balance of accuracy, cost, and reliability. Appropriate motor control and a simple mechanical design are helpful in producing clear and readable Braille output from different types of input.

6.1 MECHANICAL DESIGN

The mechanical design of the Braille embosser gives a foundation for all the components of the Braille embosser and enables these components to function correctly. It includes the design of the motors, the guiding components, the paper holders, and the punching mechanism. This component of the Braille embosser is of critical importance since it has a direct impact on the accuracy of the Braille dots produced.

Accuracy is a fundamental need for the design of the Braille embosser. Braille has to have appropriate spacing for the dots and characters. Thus, all the components have to be controlled accurately. The guiding rails are used to ensure that the movement is in the correct path to ensure accuracy. Another important aspect of the design of the Braille embosser is strength. The design has to be strong to ensure that it does not vibrate when punching the paper repeatedly. When the design is not strong, it may cause vibrations that impact the quality of the Braille dots produced.

The ease of assembly and maintenance is also considered in the design of the Braille embosser. The components are placed in a simple manner to ensure that they are easily assembled or replaced when the need arises. The system is designed in such a way that it can be easily accessed for adjustments and maintenance. This may be important in an educational setting.

6.1.1 Braille Cell Dimensions

The dimensions of the braille cells are standardized so that the reading of the text is possible by touch. Each braille cell consists of six dots that are arranged in two columns and three rows. The space between the dots must be uniform so that the reader can differentiate between the braille cells.

In the proposed system, proper attention has been given to the maintenance of the dimensions of the braille cells. The space between the dots in a braille cell is fixed, and the space between the braille cells is also fixed so that the character is well separated and clear for reading.

For the maintenance of the dimensions of the braille cells, proper coordination is required between the motors and the punching mechanism. The horizontal movement of the print head is responsible for the space between the braille cells, and the vertical movement of the punching pin is responsible for the dots in the braille cells.

If the dimensions of the braille cells are not properly maintained, the dots may overlap and may not present a proper pattern for reading the text. The system is designed to keep the dimensions of the braille cells close to the standard braille measurements.

6.1.2 Paper Feed Mechanism

The paper feed mechanism is responsible for controlling the paper's movement during the printing process. Once a line of Braille text has been printed, the paper needs to be moved to the next position to accommodate the next line of text to be printed.

The use of stepper motors is also important because it has the ability to move in steps. This helps in controlling the movement of the paper properly so that the distance between the lines remains even. Once a line is printed, the motor moves the paper by a certain degree so that the next line comes in the correct position for printing.

The paper feed mechanism has also been set in a way that the paper does not slip while it is being printed. This has been achieved by using rollers or clamps to keep the paper tightly in place. Even if the paper slips a little while it is being printed, it will affect the placement of the dots on the paper. This makes it difficult for a blind person to read the Braille that has been printed. So, it is important that the paper does not slip while it is being printed.

The mechanism has also been set in a way that it can print more than a single page. The paper passes smoothly through the mechanism without any tearing or getting stuck. The alignment of the paper remains proper from the start to the end of the printing process.

6.1.3 Punching Assembly

The most important part of the hardware is the punching assembly, as it is the one that actually makes the Braille dots on the paper. It is made up of a small pin or needle that is pushed into the paper to make the raised dots.

The design of the punching assembly is such that it is able to apply the same force to the paper every time. If the force is not enough, the Braille dot may not be well formed. If the force is too much, the paper may tear. Thus, the system is able to apply the right force as it punches.

The mechanism for punching is made up of a servo motor, as it is able to move precisely. The motor moves the pin downwards to make the Braille dot and then moves it back to its original place. This is repeated to make all the Braille dots.

Another significant part of the hardware is the positioning mechanism. As there are three rows in one Braille cell, the pin needs to be moved vertically to the correct row before punching. This is achieved through the use of another servo motor, as it is able to position the pin well.

6.2 MOTOR CONTROL SYSTEM

The motor control system is used to control all the movements of the Braille embosser. It ensures the print head is at the right position, the paper is moving correctly, and punching is done at the right time.

In this system, stepper motors and servo motors are used. Stepper motors are used when precision is required for the movement of the parts, such as the print head and the paper. Servo motors are used for punching. The Arduino is used as the main controller for the system and sends signals to the motor based on the command received from the software.

The coordination of the stepper and servo motors is of prime importance for the proper working of the system. Each step of the process should be done in the right order. For instance, the position of the print head should be reached first, then punching should be done, and only then should the paper move. If these steps are not done in the right order, then the output may not be received correctly. Sometimes a small mistake in the timing of the steps can also lead to improper alignment of the parts involved.

6.2.1 Block Diagram

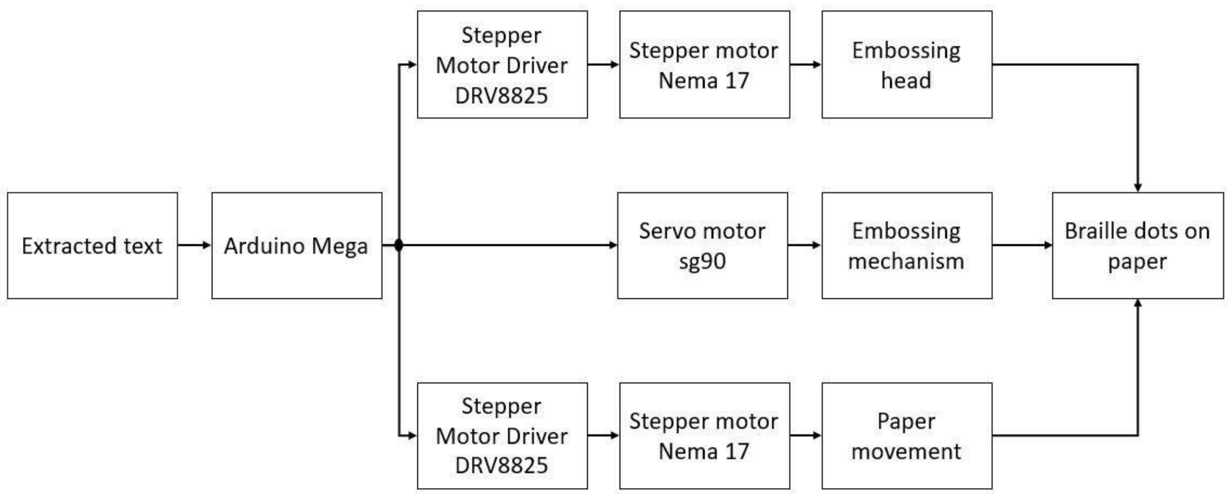

Fig 6.1: Hardware Block Diagram of Braille Embosser System

The block diagram depicts the working and the interaction between the various hardware components of the Braille embosser system. The extracted text from the software section is given as an input to the Arduino Mega microcontroller. This acts as the main control unit. The Arduino receives the input data and processes it. This processed information is given as a control signal to the stepper motors. The stepper motors are driven by the stepper motor drivers (DRV8825), which control the NEMA 17 stepper motors. One stepper motor is responsible for the horizontal movement of the embossing head, while the other stepper motor is responsible for the vertical movement of the paper.

In addition to the stepper motors, the Arduino also gives PWM signals to the servo motors (SG90), which are responsible for the punching action. One servo motor is responsible for the positioning of the punching pin at various rows of the Braille cell, while the other servo motor is responsible for the punching action.

The working of the embossing head results in the creation of Braille dots on the paper. The final output is the creation of Braille characters.

6.2.2 Circuit Diagram

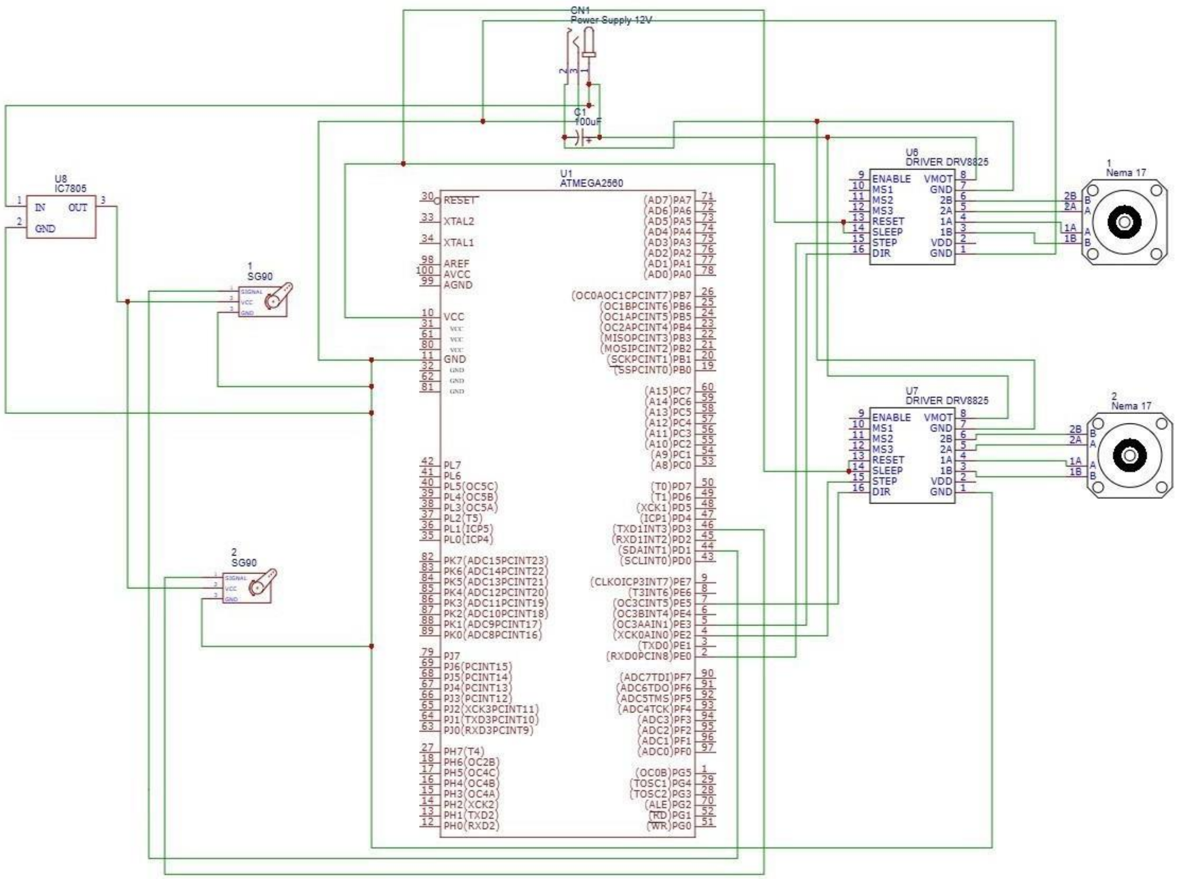

Fig 6.2: Hardware Circuit Diagram of Braille Embosser System

The circuit diagram for the proposed system shows the interconnection between the microcontroller, motor drivers, motors, and the power supply components. The Arduino Mega functions as the main controlling component for the system by receiving the processed information from the software.

There are two motor drivers connected to the system for the stepper motors. The stepper motors are responsible for the horizontal and vertical movements of the system. The stepper motors are connected to the motor drivers, which provide the necessary current for the stepper motors.

There are two SG90 servo motors connected to the system for the positioning and punching functions. The servo motors are connected to the Arduino for the necessary control signals.

A regulated power supply is connected to the system for providing the necessary power to the components. The 7805 voltage regulator is connected to providing a constant 5V to the components. A constant 12V is provided to the stepper motors through the motor drivers.

This circuit diagram for the proposed system shows the interconnection between the components for the Braille embossing system. This interconnection between the components will enable the Braille embossing system to work efficiently.

6.2.3 NEMA 17 Stepper motor

Fig 6.3: Stepper motor

NEMA 17 stepper motors are utilized to control both horizontal and vertical movements in this system. This type of motor is recognized for its accuracy and reliability, and it is often utilized in various applications where precise control is required.

One of the stepper motors is utilized to control the print head's horizontal movement. This is crucial in enabling the system to move horizontally to any character position in a line of text. The other stepper motor is utilized to control vertical movements of the paper, enabling line printing.

The stepper motors are utilized in this system in conjunction with drivers that control current flow to the motors and control signals from the Arduino system. This ensures that the motors' speeds and directions are controlled by the system.

The utilization of NEMA 17 stepper motors ensures that there is consistent and precise movement in this system, which is crucial in ensuring precise spacing between dots and characters.

6.2.4 SG90 Servo Positioning

Fig 6.4: Servo motor

SG90 servo motors are utilized for positioning and punching actions. These servo motors are small in size and affordable, with high accuracy in angular movements.

One servo motor is utilized for positioning the punching pin in a vertical position. The servo motor positions the punching pin in various rows in the Braille cell. This enables the creation of dots in all six positions.

The second servo motor is utilized for the punching action. The servo motor punches the dots using the pin in a downward position. After punching the dots, the servo motor returns to its initial position. The utilization of servo motors enables the system to perform movements with high accuracy and efficiency due to the high response time of servo motors.

6.2.5 CNC Shield

A CNC shield is another interface board that can be used to increase the capacity of the Arduino Mega 2560 board to control several stepper motors at the same time. The CNC shield is basically a board that bridges the Arduino board and the motor driver, allowing the motor driver to have sockets where motor driver modules, such as A4988 and DRV8825, can be connected. The CNC shield also helps to organize the connections of the motor by providing the necessary connections between the Arduino board and the motor driver, thereby eliminating the need to connect the motor to the Arduino board in a complex manner. The CNC shield also has the capacity to control the movement of the motor along the X, Y, and Z axes.

6.2.6 Stepper Motor Driver (A4988/DRV8825)

A stepper motor driver is a significant part of the project, which makes the control of the NEMA 17 Stepper Motor possible with a high degree of accuracy. It works on the principle of receiving low-power digital signals, i.e., steps and direction, from the Arduino board and then transforming these signals into controlled electrical currents, which are then used for driving the motor. The motor driver energizes the motor coils in a particular sequence, which makes the motor rotate in steps with a high degree of accuracy. In addition, the motor driver also offers the facility of micro-stepping, which makes the motor movement smooth and makes the motor suitable for use in applications where a high degree of accuracy and stability are required. This precise control is essential in the Braille embosser to ensure correct positioning of the print head and accurate formation of Braille characters.

6.2.7 Linear guide rods

Linear guide rods are used for giving a straight path for the smooth movement of the carriage. These rods are generally made of materials such as stainless steel or coated steel so that they are strong and tend to last for a longer period of time without getting worn out easily.

The rods are also helpful in preventing any kind of side movement of the rods, which is helpful in keeping the embossing head correctly aligned over the paper. This is important for the correct placement of Braille dots over the paper. Since the rods are strong and tend to last for a longer period of time without getting worn out easily, they are helpful for the motors to move the carriage correctly and produce clear Braille dots over the paper.

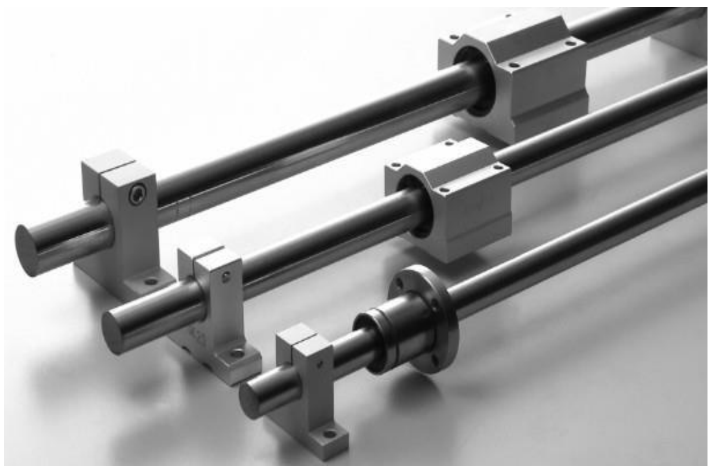

Fig 6.5: Guide rods

6.3 POWER REQUIREMENTS

The system utilizes a regulated DC voltage supply for the smooth functioning of the control and actuating devices. In general cases, the Arduino board is supplied with a voltage of 5V. It is regulated by a voltage regulator, such as a 7805 IC. It is essential to maintain a regulated voltage to avoid any fluctuation that might impact the performance of the microcontroller. In addition to this, devices such as stepper motors and servo motors require higher voltages. To meet this requirement, a separate voltage supply is arranged for the motors. It is ensured that the Arduino board is not overloaded with the voltage supply. In addition to this, the system also ensures that there is proper grounding of all the devices to ensure smooth communication between the devices.

In addition, the distribution of power within the system is well planned in order to enhance efficiency and safety. The capacitors can be employed in filtering out the noise and regulating the voltage supply during sudden changes in the load, especially during the start and stop of the motors. The dissipation of heat is another aspect that is taken into account, especially in the voltage regulators and motor drivers, in order to avoid overheating during constant use. The use of proper insulation, wiring, and fuses can help in protecting the system from short circuits, which may cause damage. All these aspects ensure the proper and long-lasting use of the Braille embosser.

Chapter 7

BRAILLE CONVERSION ALGORITHM

The algorithm for conversion to Braille would be one of the most important aspects of the proposed system. This is so because this algorithm would be used to convert the extracted textual data into a tactile form so that it can be easily readable by visually impaired persons. Even though the previous stages of the system would be used to extract and collect textual data from various formats and forms of input media, this stage would be used to ensure that the extracted textual data is correctly represented in Braille form.

However, the Braille is not just a direct translation of the print. Instead, there are some rules to be followed in the process. This not only makes the text easier to read in Braille, but also makes the text shorter. Therefore, the process of converting the normal text to Braille is not just about replacing the alphabets with the dot patterns. Instead, it is also about the formation of words, the use of contractions, and the correct use of numbers or symbols.

In the present project, Grade 2 Unified English Braille is used for the conversion of the text. This type of Braille is mostly used in the classroom setting. This type of Braille also includes contractions for the most commonly used words and combinations of alphabets. This makes the text shorter and easier to read. This type of Braille also includes the use of the Nemeth code for the inclusion of mathematical and technical materials. This makes the system not only suitable for the normal text, but also for the academic and study-related materials.

Once the Braille has been created, the next step is to convert it into digital form that the hardware can recognize. This is accomplished by using a form of binary code, where the presence of dots on the Braille corresponds to the presence of the number one, and the absence of dots corresponds to the presence of the number zero. The reason that the embossing process has to be mirrored is that the punching action of the machine is downward, so the result has to be flipped.If the mirroring does not happen, then the Braille will appear to be reversed and will not be easy to read. This step is important to ensure that the output is correct.

Finally, the data is formatted according to predetermined standards for page formatting. This is important to ensure that the data is aligned correctly and remains readable. All of the steps of this algorithm are carefully constructed to ensure that the result is accurate, consistent, and useful.

7.1 GRADE 2 UEB TRANSLATION

Unified English Braille (UEB) Grade 2 is used in this system for the conversion of textual information to Braille form. Grade 1 Braille uses individual characters for letters, whereas Grade 2 uses contractions, which help in reducing the number of characters used in the text. This makes the text shorter and easier to read, which is another advantage in the increased speed of reading. The use of Grade 2 Braille in this system is beneficial because it can convey more text in less space. This is beneficial for printing as well as for the reader, as it saves space and the text is manageable.

At this point, the system receives the text that was extracted and checks for any pattern that can be replaced using Grade 2 contractions. The system should recognize these correctly and use the contractions correctly without affecting the meaning of the text. The system should also check where the contraction should be applied and where it should not be applied. Some of the contractions depend on the word’s position, so caution should be taken while applying the contraction. If the contraction is applied incorrectly, it will confuse the reader. Therefore, this step is carried out with caution in order to produce text that is clear and easy to read. However, the process of translation is not as simple as it seems, as some words may require context-based contractions. For instance, there are words that need to be contracted only when they are used at the start or end of a sentence or a word.

Thus, it is important that the system makes use of these rules in order to ensure that there are no translation errors and that the meaning of the original text is maintained. The use of Grade 2 Braille rules will improve the readability of the output of the system and make it more efficient in terms of embossing the output. This will make the system more practical for use in the future, especially for those who are already familiar with contracted Braille.

Punctuations in Braille:

Punctuations	Print Symbols	Braille Dots
apostrophe	'	⠄
oblique stroke	/	⠌
Square bracket opening	[	⠷
Square bracket closing	]	⠾
colon	:	⠒
Comma	,	⠂
dash	--	⠤⠤
double dash	----	⠤⠤⠤⠤
ellipsis	...	⠄⠄⠄
exclamation mark	!	⠖
hyphen	-	⠤
parenthesis, opening	(	⠶
parenthesis, closing	)	⠶
period/fullstop	.	⠲
question mark	?	⠦
quotation mark, double opening	“	⠦
quotation mark, double closing	”	⠴
quotation mark, inner opening	‘	⠠⠦

Fig 7.1: Braille Representation of Common Punctuation Symbols

7.1.1 Contraction Rules

The contraction rules are an essential part of the Grade 2 Braille. These rules help in the formation of common words or letter combinations using fewer Braille cells, making the text shorter and easier to read. In this system, the contraction rules are implemented using a lookup table. In this method, the system scans the input text, compares it with the predefined rules, and replaces the word or letter combination with the corresponding Braille symbol if it finds a match.

For instance, the common words “and,” “for,” “with” can be replaced using the corresponding Braille symbol. Similarly, letter combinations such as “ch,” “sh,” “th” can also be replaced using the corresponding Braille symbol. This helps in making the text shorter, thus enhancing the reading speed.

While applying the above-mentioned rules, care is taken to ensure the correct usage of the rules. In the wrong usage of the rule, the meaning of the text can be changed, causing confusion. The system also takes care of situations where more than one rule can be applied to the input word. In such cases, the system decides the rule to be applied based on priority, thus avoiding mistakes and keeping the output clear.

7.1.2 Strong/Weak Contractions

In Grade 2 Braille, the use of contractions is classified into strong or weak, based on the usage and importance of the word. Strong contractions represent the complete word, and they are often used for common words such as “and,” “the,” and “with.” In this regard, the writer does not have to write the word letter by letter, but instead, the whole word is represented by one Braille symbol. This makes the writing shorter, thus easier to read. It also enhances the reading speed, especially when dealing with long texts. The use of strong contractions also plays an important role in saving space on the paper, especially when printing the Braille. At the same time, it makes the text uniform, thus easier to read for the person who reads it. This makes the entire system efficient, thus easier to use in the real world. It also ensures the reading process is smooth, thus less tiresome for the person who reads it.

On the other hand, weak contractions are the parts of words or common letter combinations like “ing,” “ed,” “er,” etc. These are applied to the words in the text to shorten them without changing the original meaning of the text. Unlike strong contractions, weak contractions are applicable under certain conditions like the position of the word and the combination of letters. This makes the application of weak contractions slightly complex, as the system has to identify the pattern of the weak contractions and apply them according to the conditions.

In this system, the differentiation between strong and weak contractions is of major importance and has to be done correctly in order to apply the rules of contraction correctly. In the case of strong contractions, the rules are applied if the entire word matches the pattern of the strong contraction. In the case of weak contractions, the pattern matching has to be done correctly, and the conditions have to be satisfied. The system has to apply the rules of contraction in the correct manner if there are multiple rules applicable to the text. If the rules are applied incorrectly, it may cause ambiguity in the text and make the reader confused. Hence, the proper handling of rules is required to maintain clarity in the Braille output.

7.2 NEMETH MATH CODE MAPPING

Nemeth code is a code used for representing mathematical and technical content in Braille form. Unlike other texts, mathematical content must be accurately represented in terms of symbols, numbers, and operations.

In this process, Nemeth code is applied for the content identified as numerical or mathematical in nature in the input text. This includes numbers, mathematical operations, and equations. The process accurately identifies these mathematical expressions in the text and represents them in Braille form according to the Nemeth code. Another important aspect of the Nemeth code is the use of special indicators in Braille form. A number indicator is used to differentiate numbers from letters. In mathematical content, addition, subtraction, and multiplication operations are represented in Braille form. Processing mathematical content also includes maintaining the original order and structure of the content. This includes accurately representing mathematical equations in Braille form. This stage adds a layer of complexity in the process, but it enhances the overall ability of the system significantly. This makes the process useful for educational purposes, as the embosser can be used for mathematical content as well.

Print Number Symbols	Braille Number Symbols
1.	⠼⠁
2.	⠼⠃
3.	⠼⠉
4.	⠼⠙
5.	⠼⠑
6.	⠼⠋
7.	⠼⠛
8.	⠼⠓
9.	⠼⠊
0.	⠼⠚

Fig 7.2: Braille Representation of Numerical Symbols

7.3 BINARY GENERATION AND MIRRORING

Once the text has been converted into the Braille pattern, the next step is to convert this pattern into a form that the hardware system can recognize and work with. This is achieved by the use of binary encoding.

Each of the Braille cells has six dots, and the status of each of these dots is given by the binary number. A value of 1 represents the presence of the dot, whereas a value of 0 represents the absence of the dot. By combining these values, the Braille character can be given a unique binary representation.

The binary representation of the Braille character is then used to control the embossing mechanism. The individual values are given to the hardware, which uses this information to determine the status of the individual dots.

However, as the hardware uses the downward punching method, the binary representation of the Braille character has to be mirrored before it is given to the hardware. This is to ensure that the output appears correctly when viewed from the front side of the paper.

7.3.1 6-Dot Cell Encoding

The basic unit of the Braille alphabet, known as the Braille cell, contains six dots that are organized into a matrix of three rows and two columns. Each of the positions of the dots has a corresponding number.

In the encoding scheme, the positions of the dots are given corresponding positions in the binary number system. This allows the data to be represented in the form of binary numbers, where if certain dots are present, the corresponding positions are given the value of 1, and the rest are given the value of 0. This method of encoding has the advantage of providing a simple way of representing the data, which can be easily processed by the hardware and the software.

It is important to note that consistency is required in the encoding scheme to ensure that the hardware is able to correctly interpret the data that has been fed into it. Any errors in the encoding scheme can result in incorrect dot patterns, which may affect the readability of the data.

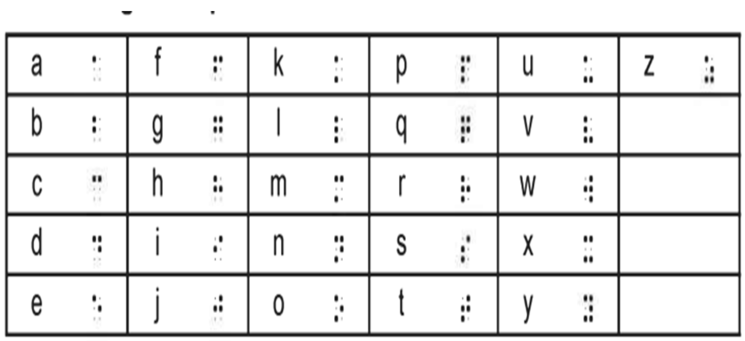

Fig 7.3: Standard Braille Alphabet Representation

7.3.2 Downward Punch Mirroring

The mechanism of embossing, which is employed in this system, consists of punching Braille dots onto the paper from the back. This, in turn, causes the output to appear reversed when viewed from the front of the paper.

In order to correct this, the system carries out a mirroring operation on the binary data before it is sent for processing. This is done by reversing the order of bits in the binary pattern. This is required for ensuring that the output is correctly oriented for use.

The mirroring operation is critical for ensuring that the output of the system is easily readable. If this operation is not performed, it would be difficult to read the embossed text, as it would not match the expected pattern. The mirroring operation is performed on all Braille cells.

7.4 PAGE FORMATTING STANDARDS

Page formatting is one of the important steps in Braille conversion, and this step is essential to make sure that the final output is clear and easy to read. With the Braille format, the page arrangement is determined by fixed limits such as the number of characters in a line and the number of lines in a page. This is determined by the size of the paper and the way the machine works. This ensures that there is no overflow of the text and everything is well arranged. In addition, it ensures that the spacing is well maintained throughout the page, which is essential for easier reading.

The next aspect is the spacing and the way the lines are broken. This is also well addressed by the system. The system ensures that the words are not broken in the middle and that there is sufficient spacing between the words and the characters. This ensures that the person reading the Braille is not confused. In addition, it ensures the smooth movement of the machine parts, as the printing is done in a fixed manner. This ensures the printing is neat and easy to follow. This also ensures that there is no error in the printing and the output is as required.

Chapter 8

RESULT OF PHASE – 2 IMPLEMENTATION

In this stage, all the different components of both the software and hardware of the Braille embosser are integrated and made to work together in a single entity. The integration of the components is then tested with different types of input to ensure that it is able to read the input correctly and then convert it to Braille before embossing it on a piece of paper. This stage is important in ensuring that all the components are working correctly and that the entire process is running smoothly without any problems. The main goal of this stage is to ensure that the entire system is able to produce accurate Braille output that is clear and reliable.

8.1 INPUT PROCESSING AND TEXT EXTRACTION

During this phase, the system was tested with different input formats such as images, PDFs, and presentation files. In the case of text extraction, the system was able to perform well with the use of Google Gemini. In the case of images, the system was able to perform well with very little error. In the case of PDFs, each page was first converted into an image and then extracted one by one. In the case of presentation files, the text was extracted directly from the presentation. In some instances where the API was not responding well, EasyOCR was also used as a fallback option. In all instances, the system was able to work well without any interruption.

The system was able to handle different font styles and sizes without major difficulties. However, there was some dependency on the quality of the input data, particularly where images of handwritten texts were used. In spite of all this, the system was able to maintain consistency by the use of EasyOCR as a fallback option. This phase was also able to prove that the system can work well under different conditions. It also proved that the approach being used is effective. The system was able to perform well even when the test was carried out repeatedly, thus ensuring that the results were consistent. This gives one the confidence that the system can perform well in real-world applications, as it does not fail repeatedly.

Table 2: OCR accuracy

TEXT EXTRACTION (OCR) ACCURACY REPORT

IMAGE NAME	ACCURACY (%)
image1.jpg	100.00%
image2.jpeg	93.57%
image3.jpg	100.00%
image4.jpg	100.00%
image5.jpeg	97.83%
image6.jpeg	99.53%
image7.jpeg	98.82%
image8.jpeg	95.97%
image9.jpeg	97.78%
image10.jpeg	100.00%
image11.jpeg	98.65%
image13.jpeg	87.76%
image14.jpeg	70.44%
image15.jpeg	83.15%
image16.jpeg	8.76%
image17.jpeg	0.79%
image18.jpeg	100.00%
image19.jpeg	100.00%
image20.jpeg	100.00%
image31.jpg	100.00%
image32.jpg	96.77%
image33.jpg	98.11%
image34.jpg	97.14%
image35.jpg	100.00%
image36.jpg	100.00%
image37.jpg	95.65%
image38.jpg	97.44%
image39.jpg	94.12%
image40.jpg	100.00%

OVERALL OCR ACCURACY: 90.08%

8.2 SERIAL COMMUNICATION AND DATA TRANSFER

The communication between the software and the hardware was established by using serial communication (UART) at a baud rate of 9600. The processed text was sent line by line from the software to the Arduino Mega 2560.

A handshake mechanism was incorporated to facilitate the synchronization between the software and the hardware. The software sends each line of the processed text, and then it waits for a "Ready" message from the Arduino Mega 2560 before sending the next line of the processed text. This ensures that the hardware has finished embossing the line of the processed text before it receives the next line of the processed text.

8.3 BRAILLE CONVERSION

The Arduino device was successful in converting the received text into Braille patterns according to the predefined mapping. The device can perform alphabet mapping, Grade 2 contractions, and the representation of numbers using a number indicator, as well as simple mathematical signs.

Each character is translated into a 6-dot Braille cell, which is encoded as a binary number. The 6-dot Braille cells can then be used to control the embossing process. The contraction rules help to reduce the number of Braille cells, which can save space.

Table 3: Braille translation accuracy

BRAILLE TRANSLATION (LOGIC) ACCURACY

IMAGE NAME	LOGIC ACCURACY (%)
image1.jpg	100.00%
image2.jpeg	98.15%
image3.jpg	100.00%
image4.jpg	100.00%
image5.jpeg	98.90%
image6.jpeg	99.20%
image7.jpeg	100.00%
image8.jpeg	97.45%
image9.jpeg	96.80%
image10.jpeg	100.00%
image11.jpeg	99.10%
image13.jpeg	95.40%
image14.jpeg	96.20%
image15.jpeg	94.85%
image16.jpeg	98.50%
image17.jpeg	97.20%
image18.jpeg	100.00%
image19.jpeg	100.00%
image20.jpeg	100.00%
image31.jpg	100.00%
image32.jpg	98.70%
image33.jpg	99.10%
image34.jpg	98.45%
image35.jpg	100.00%
image36.jpg	100.00%
image37.jpg	97.80%
image38.jpg	98.30%
image39.jpg	96.90%
image40.jpg	100.00%

OVERALL LOGIC ACCURACY: 98.65%

8.4 MOTOR CONTROL AND MECHANICAL OPERATION

The hardware system of the Braille embosser was designed to ensure precise and accurate movements in order to emboss Braille characters onto the paper. The hardware system consists of NEMA 17 stepper motors, which are used for controlling both the carriage movement (X-axis) and the paper feeding (Y-axis) system. The stepper motors operate in precise steps, enabling the printer head to come to rest at an exact location where it is required for each Braille cell. The precise location is necessary for ensuring proper spacing between Braille dots.

The embossing is done by the use of SG90 micro servo motors. The micro servo motor is used for vertical positioning, enabling the punching nail to reach the required row in the Braille cell. The required rows include the top, middle, and bottom. The second micro servo motor is used for punching, where it is responsible for pushing the punching nail onto the paper. The use of two micro servo motors ensures precise accuracy in the creation of Braille dots. To ensure consistency and avoid errors, the system also has a homing feature that makes use of a limit switch. After printing a line of Braille, the printer head is reset back to the original position using the limit switch. This ensures that the printing of the next line of Braille starts in the correct position.

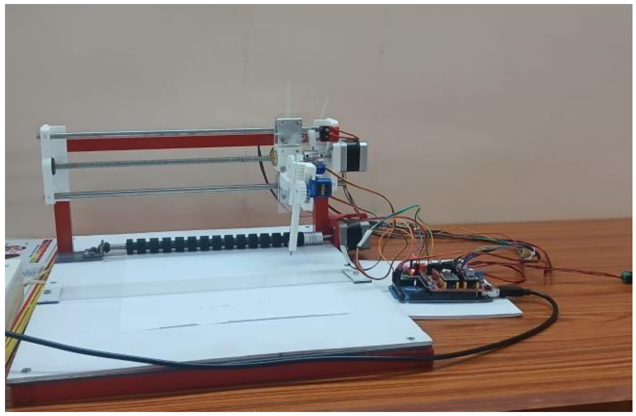

Fig 8.1(a): Braille Embosser

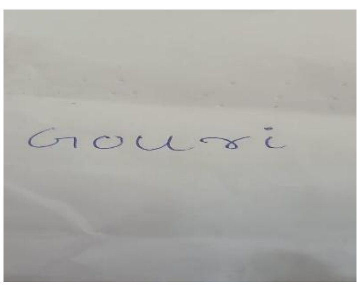

Fig 8.1(b): Input

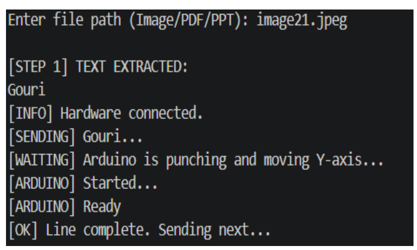

Fig 8.1(c): Extracted text

Fig 8.1(d): Braille output

8.5 SYSTEM PERFORMANCE

During the test, the Braille embosser system showed consistent and reliable results. The coordination between the software and hardware was seamless, and the communication protocol, specifically the handshake mechanism, was effective in ensuring that the software was sending the right information that was correctly interpreted by the hardware without any loss of information or errors in processing the information.

The processing time was also observed to vary depending on the length of the input text and the number of Braille dots to be embossed. However, despite the slow processing speed, the Braille embosser system was able to provide precise results, which is essential in the production of Braille since the visually impaired would be the ones to read the produced Braille.

Based on the overall performance of the Braille embosser system, it may be concluded that the system may be considered effective in the development of assistive technology, even though it may not be as fast as the industrial version of the embosser in the market, since the system was also able to provide precise results despite the slow processing speed, thus proving that a software and hardware combination may be used in the development of assistive technology that provides reliable results.

Table 4: System Performance

Category	Performance
Extraction Accuracy	90.08%
Logic Accuracy	98.65%
Internal Processing Delay	~5 seconds
Character Printing Speed	**~9.6 Seconds** per character (worst-case, full 6-dot cell)
Timing Delay	**1.3 Seconds** per dot

Chapter 9

CONCLUSION

The proposed Braille embosser system is a complete solution for converting various types of input into Braille and then producing a physical output through embossing. The major aim of this project is to develop a system that is not only cost-effective but also efficient and easy to use so that it can be of use to visually impaired people. This aim of the project has been achieved in a successful manner by integrating both software and hardware aspects of the system in a practical manner. The major advantage of this system is its ability to work with various types of input such as handwritten images, PDF files, and PowerPoint presentations. This is a major advantage of this system as it is easy to use and practical due to the elimination of the need for various tools to be used for different types of input files. Regarding the software aspect, the system is focused on extracting the text and converting it to Braille format. An AI-based approach is primarily used for extracting the text, which helps in the identification of printed and handwritten texts. In case the AI-based approach does not produce accurate results, the use of an OCR-based approach is considered for better results. Once the text is extracted, it is converted to Braille format using encoding techniques. Grade 2 Braille is used for normal text to save space by using contractions, while Nemeth Code is used for mathematical expressions.

Regarding the hardware aspect of the system, it is focused on ensuring precise movement of the paper and embossing of the Braille dots. Stepper motors are used for the horizontal movement of the print head and the vertical movement of the paper. On the other hand, servo motors are used for the positioning of the paper and punching of the Braille dots. One servo motor is used for the positioning of the pin, while another servo motor is used for punching the dots. The entire system is line-based, which means that a Braille character is correctly formed before moving to the next character. The system was tested with different types of inputs to check the performance of the system. These tests were helpful in validating that the system could accurately extract the text and produce readable Braille output.

There are a few limitations of the proposed system. The accuracy of the extracted text also depends on the quality of the input, especially when it comes to handwritten input. The speed of the printer is also relatively low since it is a sequential operation. Additionally, the proposed system only provides English Braille; therefore, it could be extended to include more languages in the future.

In conclusion, the proposed system provides a viable solution for those who need a Braille embosser but want a low-cost solution using readily available components such as microcontrollers, stepper motors, and servo motors. The proposed system has successfully demonstrated how software and hardware can be used together to produce a viable solution for the visually impaired.

REFERENCES

[1] Jonnatan Arroyo, Ramiro Velazquez, Mehdi Boukallel, Nicola Ivan Giannoccaro ,Paolo Visconti, "Design and Implementation of a Low-Cost Printer Head for Embossing Braille Dots on paper", *IJETER*, vol 8.No.9, 2020, p:8.

[2] Abdullah Farhan Siddiqui, Md Misbahuddin, "A Tool for Converting Text to Braille Script to Empower Readability for Blind People",*JECNAM*,2799-1156,Vol: 04, No.03,April-May 2024 https://journal.hmjournals.com/index.php/JECNAMDOI: https://doi.org/10.55529/jecnam.43.1.7

[3] Andrean Ignasius, Joerio Christo Chandra, Roberto Oscadinata, Derwin Suhartono, "Image Pre-Processing Effect on OCR's Performance for Image Conversion to Braille Unicode",8th *ICCSCI*,2023

[4] Falgoon Sen Apu, Fatema Islam Joyti, Md Ala uddin Anik, Md Wasi Uddin Zobayer, Atanu Kumar Dey, Sakib Sakhawat, "Text and Voice to Braille Translator for Blind People", *IEEE*, ACMI,8-9 July 2021

[5] Sangeeta Kumari, Akshay Akole, Pallavi Angnani, Yash Bhamare, Zaid Naikwadi, "Enhanced braille display use of OCR and solenoid to improve text to braille conversion",*IEEE*,2020 INCET,2020, p:5